SOUVIK PODDAR

Mechanical Engineer | Electromechanical Systems | Product Development

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SUMMARY

PhD in Mechanical Engineering with 7+ years of experience developing and validating electromechanical systems. Skilled in CAD, mechatronic integration, control, rapid prototyping, FEA-based design validation, and biomechanics. Proven cross-functional execution across mechanical, electrical, controls, and experimental workflows.

CORE SKILLS

- Mechanical design: Component/assembly design, electromechanical subsystem design, mechanism/fixture design
- Hardware Integration: Motor drives, absolute/incremental encoders, sensors, wiring/harnessing, safety interlocks, modular interfaces
- Controls: Closed-loop control (PID), DAQ, signal processing, multimodal synchronization
- Prototyping & testing: Test fixtures, benchtop validation, rapid prototyping, machining, 3D printing, laser cutting
- Simulation & Analysis: FEA-driven design validation, nonlinear modeling, Python scripting for FEA
- Biomechanics: Gait analysis, kinematics, motion capture, experimental protocols

TECHNICAL TOOLS

- CAD: SolidWorks, Creo, CATIA, Inventor
- FEA: Abaqus/Explicit, ANSYS
- Controls: LabVIEW, Embedded C
- Programming and Data Analysis: Python, MATLAB/Simulink, SPSS
- Motion Capture: Vicon Nexus, OptiTrack Motive, Motion Analysis Cortex

EXPERIENCE

Research Assistant, University at Buffalo **Buffalo, NY** (Aug 2020 - Present)

- Designed and delivered I-PEDLE (Intelligent Prosthesis Emulator for Daily Living Enhancement), a 380 g, 3-DOF robotic platform for evaluating upper-limb prosthesis configurations via adjustable weight and functional modes.
  - Developed CAD models from concept to release: component design, assemblies, electronics mounting, and routing clearances.
  - Engineered a plug-and-play electromechanical architecture integrating NI myRIO controllers, encoders, Hall sensors, motors, load cells, and joystick I/O with serviceable harnessing in a compact structure.
  - Implemented closed-loop motion control and real-time sensing pipelines. Validated via step-response and trajectory-tracking tests (up to 0.7 Hz, under 5% tracking error, no overshoot).
  - Iterated hardware using 3D printing and machining. Built fixtures and test rigs for repeatable verification.
  - Conducted a controlled user study (n=10) across four wrist configurations, quantifying design trade-offs via task performance and upper-limb kinematics.
- Implemented robot-resisted gait training with biofeedback for adults with cerebral palsy (n=5). Improved step length (+9.46%), walking speed (+10.50%), body posture, muscle activation.
  - Led a 4-person cross-functional team and managed robot control and data acquisition.
  - Diagnosed and resolved intermittent runtime issues during live sessions to ensure safe, repeatable operation.
- Teaching Assistant (Aug 2020 - May 2022): Road Vehicle Dynamics, Manufacturing Processes, Intro to ME/AE Practice.
- Mentored master's and undergraduate students on research projects and deliverables.

Visiting Scholar, University of Michigan **Ann Arbor, MI** (Jan 2025 - Jan 2026)

- Led an I-PEDLE user study (n=15), systematically varying prosthesis weight and quantifying effects on task performance, upper-limb kinematics, and perceived workload (NASA-TLX).
- Designed and 3D-printed study fixtures, built the experimental setup, and developed the end-to-end protocol and data-collection workflow.
- Translated study results into design recommendations by identifying trade-offs between prosthesis weight, functional performance, user workload and fatigue.

Research Assistant, Arizona State University **Tempe, AZ** (Aug 2018 - May 2020)

- Modeled and optimized inflatable actuator composites using FEA sweeps (33/66/100% volume) and benchtop validation. Achieved ~4.5x faster pressurization.
- Contributed to building a portable pneumatic source and control system (0.131 MPa max pressure, 21.45 SLPM max flow) for wearable-robot operation.
- Designed and characterized a fabric-based soft ankle module to emulate ankle stiffness. Reached ~50% of biological ankle stiffness.
- Developed inflatable insole sensors for gait-phase detection. Fabricated and validated load response up to 700 N.
- Automated Python-driven FEA simulations across soft actuator modes (stiffening, contracting, elongating). Validated via benchtop tests.

EDUCATION

University at Buffalo **Buffalo, NY**
Doctor of Philosophy, Mechanical Engineering, GPA: 3.80/4.00 May 2026

Arizona State University **Tempe, AZ**
Master of Science, Mechanical Engineering, GPA: 3.72/4.00 May 2020

National Institute of Technology **Agartala, India**
Bachelor of Technology, Production Engineering, GPA: 8.34/10 May 2018

SELECTED PUBLICATIONS

- Poddar et al. (2026). A Robotic Emulator for User-Driven Design of Multi-DOF Transradial Prostheses. IEEE Robotics and Automation Letters. DOI: 10.1109/LRA.2026.3664529
- Poddar et al. (2025). Improving Foot-Rocker via Robot-Resisted Gait Training with Self-awareness Biofeedback in Adults with Cerebral Palsy. IEEE Transactions on Neural Systems and Rehabilitation Engineering. DOI: 10.1109/TNSRE.2025.3636432
- Poddar et al. (2022). A lightweight transradial prosthetic emulator for optimizing prosthetic wrist design. IEEE International Conference on Biomedical Robotics and Biomechatronics. DOI: 10.1109/BioRob52689.2022.9925286
- All publications: https://scholar.google.com/citations?user=HyodwH8AAAAJ&hl=en&oi=ao

PATENTS, AWARDS AND FELLOWSHIPS

- U.S. Patent US 12,458,556 B2 (Issued 2025): Low-volume inflatable actuator composites for wearable applications and portable pneumatic source.
- Research supported by the NSF Disability and Rehabilitation Engineering (DARE) program award (Aug 2022 - Present).
- Master’s Opportunity for Research in Engineering (MORE) award (Aug 2019 - May 2020).